Document Intelligence

Grounded Q&A over commercial contracts

Track D -- RAG / LLM Knowledge Systems

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L1 numbers: `results/README.md`. L2 numbers: `results/gpu_default_dev_cd2a4652f434_L2/`.

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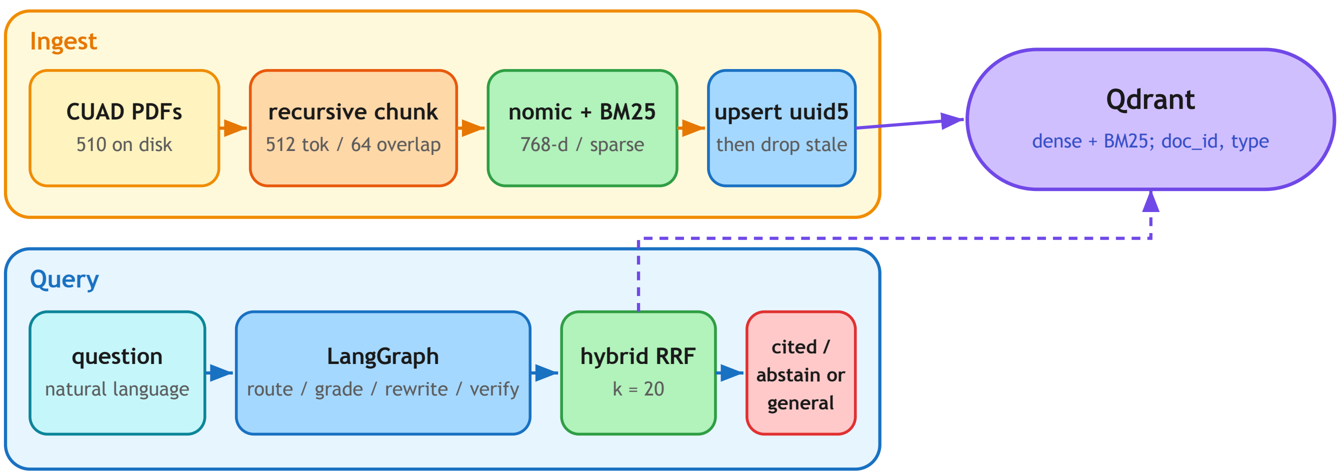
1. Problem and domain

Legal and operations teams sit on hundreds of SEC-filed commercial contracts and need clause-level answers -- governing law, termination, exclusivity, fee waivers -- with a page cite. A fluent paragraph without evidence is a liability, not a feature. CUAD v1 (The Atticus Project, CC BY 4.0) is the right corpus: 510 contracts, 41 lawyer-labelled clause types, gold spans. I index 400 PDFs, stratified over 25 agreement types. The eval dataset is 40 items over 30 of those documents.

This is a retrieval problem, not a classification or fine-tune. The labels are extractive spans, not instruction pairs; a 7B LoRA (Track B) would memorize clause phrasing and still invent cites. A classical classifier (Track C) can tag "has a non-compete" but cannot quote the trigger. A three-agent CrewAI graph (Track A) adds messages without adding evidence. Track D is the fit: retrieve the span, generate only from it, abstain when it is missing.

Figure 1. Query and ingest share one Qdrant collection. The graph decides whether to retrieve.



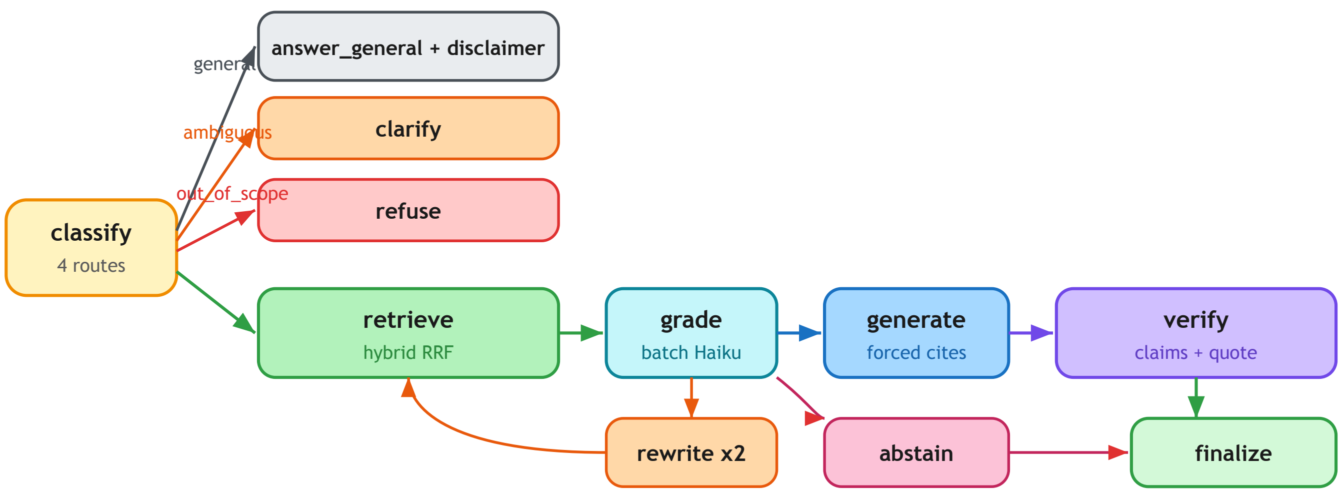
2. Approach and algorithm decisions

Every default below beat a named alternative on this corpus or was rejected for a documented reason. Stages are YAML strategies; swapping embedder or fusion does not touch graph code.

Choice	Default	Rejected, and why
Chunking	Recursive 512 / 64. Page + bboxes on every chunk.	**Page-as-chunk** mixes recitals with the operative clause. **Fixed 512** splits mid-sentence. **Late chunking** needs a long-context embedder we do not serve. A 600-token indemnity straddles two chunks; 64-token overlap plus a type header recovers it. Gold is PDF text, never CUAD `answer_start`.
Embedder	nomic-embed-text-v1.5, 768-d, local, task prefixes.	**OpenAI 3-small**: clause text leaves the machine. **bge-small**: 384-d under-resolves long legal sentences (CPU smoke only). **bge-m3**: multilingual capacity we do not need. Nomic prefixes match how the questions are phrased.

Choice	Default	Rejected, and why
Store	Embedded Qdrant, dense + sparse, payload on `doc_id` / type.	**FAISS**: no payload delete, so freshness is a full rebuild. **Chroma**: weak hybrid. Server mode is the same client.
Fusion	Client RRF, k=60. Reranker off on the demo path.	Hybrid lifts dense-only hit@5 by +0.20 (table below). Sparse-only wins this extractive L1 set; hybrid stays default for paraphrase ("what if NII goes negative"). **bge-reranker-v2-m3**: +0.03 hit@5, not worth a cross-encoder inside a 90 s graph. Client RRF is unit-testable without a live store.
Control	One LangGraph: Adaptive + Corrective (cap 2) + Self-RAG.	**Naive RAG** cannot abstain. **Multi-agent chat** re-encodes the same four LLM calls. Router / grader / verifier: Haiku. Generation: Sonnet.

Figure 2. Agent graph. First-pass hits are kept across rewrites. Unsupported claims get one strict regenerate, then abstain.



Jailbreak / live unknown routes to refuse. "Capital of France" routes to general plus a disclaimer.

Hallucination. Retrieved text is wrapped in `<evidence id=...>`. Generation emits `{answer, citations: [{chunk_id, quote}]}`. The verifier checks claims against cited chunks, fuzzy-matches the quote (threshold 90), and rejects any `chunk_id` that was not retrieved. Failure: one strict regenerate, then abstain.

Query analysis. Stamps.com governing law -> *corpus_technical*, cite. "What is an indemnity clause?" -> *general*, disclaimer. Two Maintenance docs in play -> *clarify*. Jailbreak / live unknowable fact -> *refuse*.

Freshness. Registry key is sha256 + `index_sig`. `--only-changed` embeds new bytes, upserts deterministic ids, counts, then deletes other hashes for that `doc_id`. A failed embed leaves the previous version queryable. UI uploads are `Unknown`; search ORs that into any type filter so a new PDF is visible without re-embedding the 400-doc index.

3. Results and error analysis

L1 is deterministic span match on the eval dataset (n=30). The eval runner never injects the gold `doc_id`; `FilterExtractor` has to find it from the question. L2 is the same questions plus the remaining route / abstain items (n=40). RAGAS faithfulness is the declared headline. DeepEval is the cross-check -- it saturates (0.966) and disagrees with RAGAS (Spearman 0.01 on n=19 overlap), so I do not swap the headline after seeing it.

L1 retrieval, eval dataset

Method	hit@5	R@10	nDCG@10	MRR
dense-only	0.533	0.692	0.444	0.351
hybrid RRF (default fusion)	0.733	0.758	0.502	0.390
hybrid + bge-reranker-v2-m3	0.767	0.795	0.551	0.443
sparse-only (L1 winner, not default)	0.800	0.770	0.606	0.544

L2 generation, gpu_default / eval dataset / cd2a4652f434

Metric	Value
RAGAS faithfulness (headline)	0.636
RAGAS context precision / recall	0.690 / 0.700
Custom groundedness / route accuracy / citation validity	0.605 / 1.00 / 1.00
Abstention precision / recall	0.40 / 0.75
Latency p50 / p95 / LLM calls per query	48 s / 67 s / 6.5
DeepEval faithfulness (cross-check, saturates)	0.966

Where it fails. Faithfulness 0.636 means about one third of generated claims are not fully supported by the retrieved chunk -- usually a hedge the verifier lets through ("the agreement appears to..."). Abstention precision 0.40: we refuse questions the gold can answer, typically when the grader wants two relevant chunks and the first pass finds one (governing-law items burn two rewrites and ~75 s). Parties gold is many two-letter fragments; span-match recall is optimistic and I report it separately. Sparse-only beating hybrid on L1 is real: this eval set is extractive. I am not hiding that. Hybrid is the production default for the paraphrase case the span metric under-counts.

4. Production and limitations

Production and the limit I would fix first. The 400-doc index is ~11k chunks; adding one contract must not re-embed the rest (content-addressed points, prepare-then-swap). At 1,000 concurrent queries the bottleneck is 6.5 LLM calls x 48 s p50 against provider RPM, not Qdrant. The graph already skips retrieval on *general*, batches the grader, and puts Haiku on the cheap roles. Scale-out is server-mode Qdrant, embed behind TEI, and a cache on `(config_hash, question)`. The 48 s p50 is a review tool, not a desk product: close it with a score-threshold first grade and a tighter rewrite cap. No OCR.